Ali Aslanbayli

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Education

New York University — **Sept. 2024 – May 2026**
Masters of Science in Computer Engineering — *Brooklyn, NY*
GPA: 3.8 / 4.0

University of South Florida — **Aug. 2020 – May 2024**
Bachelors of Science in Computer Science — *Tampa, FL*
GPA: 3.7 / 4.0, **Judy Genshaft Honors College**

Technical Skills

Programming Languages: Python, C/C++, Go, Javascript, Typescript, PostgreSQL, Mojo
Libraries/Frameworks: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, FastAPI, React, Langchain, LLamaIndex
Developer Tools: Jira, Postman, Linux, Git, GCP, AWS, Docker, MLFlow, Ray, OpenShift

Experience

GenAI Developer — **Dec. 2024 – Present**
NYU Research Technology Services — *New York, NY*

- Architected and deployed end-to-end Retrieval-Augmented Generation (RAG) pipelines using OpenWebUI and vector databases, directly improving model accuracy and reducing hallucinations for domain-specific queries.
- Engineered real-time data ingestion modules in Python that integrate Google Search APIs, increasing LLM context coverage by 40% and enabling the system to answer queries on live events.
- Built a user-facing agentic interface that automates complex data analysis, converting natural language directly into Pandas workflows to accelerate research operations.

Full-stack Developer — **Jan. 2023 – May. 2023**
Recepta App Inc. — *Remote, US*

- Owned the development of core backend microservices in Go and PostgreSQL, designing robust RESTful endpoints for user authentication and data management capable of handling production traffic.
- Shipped critical frontend features in React/TypeScript, including profile management and search, which directly supported the platform's rapid scaling to 10,000+ active users.
- Optimized the CI/CD pipeline using Docker and GitHub Actions, implementing automated testing and caching strategies like React memoization that improved deployment speed and application performance.

Projects

Pelmen Coding Agent (https://github.com/aslanbayli/pelmen) — **July 2026**

- Built a TypeScript coding agent CLI from scratch using Bun, OpenRouter, Commander, and Ink/React, implementing a streaming agent loop with tool calling for file reads, surgical edits, file creation, and shell command execution.
- Architected a modular monorepo with AI, agent, and CLI packages, adding persistent session restore, model switching, context-window tracking with summarization, and a markdown-based skills system for reusable agent behaviors.

FireLens (https://github.com/aslanbayli/firelens) — **July 2026**

- Built a local-first hybrid code retrieval engine that indexes repositories into SQLite by extracting AST-based symbols, semantic code chunks, normalized embeddings, and file metadata with incremental reindexing, content hashing, and embedding reuse.
- Architected exact, fuzzy, and semantic search pipelines with query routing, MCP-compatible structured output, a Streamlit UI, and Mojo-accelerated kernels for fuzzy scoring, vector similarity, and top-k ranking on local codebases.

MeshNet (https://github.com/aslanbayli/MeshNet) — **Dec. 2023**

- Developed a Heterogeneous Graph Neural Network (GNN) using PyTorch Geometric to identify devices in mesh networks, designing a custom node schema (Devices, Events, Services) to model complex non-Euclidean streaming relationships.
- Optimized the training pipeline via hyperparameter sweeping and weighted loss functions to handle class imbalance, achieving 90% accuracy and surpassing Nielsen's benchmark for resolving erroneous MAC addresses.

Leadership

Society of Competitive Programmers - *Founder/President* — **Aug. 2021 – Aug. 2023**

- Hosted numerous competitive coding competitions, and mentored teams to attend the world-famous ICPC contests.

International Collegiate Programming Contest (ICPC) - *Silver Medalist* — **March 2022**

- Working in a team of two other students secured a silver medal at ICPC Southeast USA solving 5 out of the 7 problems.